Design note: latent escalation between conflict and unrest episodes

Problem. In the current joint model, escalation is identified only at the shortest scale: κ is the effect of unrest in the previous four weeks in the same cell. Escalation over months and hundreds of kilometres is absorbed by the shared field and cannot be told apart from a shared episode. The proposal makes it identifiable.

Logit of each outcome. Cell s, week t, time step $k = k(t)$ of Δ weeks:

$$\begin{aligned}\operatorname{logit} P(\text{conflict}_{st} = 1) &= m^C_{st} + v^S_k(s) + v^C_k(s) = m^C_{st} + L^C_k(s),\\ \operatorname{logit} P(\text{unrest}_{st} = 1) &= m^U_{st} + \lambda\, v^S_k(s) + v^U_k(s) = m^U_{st} + L^U_k(s).\end{aligned}$$

$L^C_k(s)$ and $L^U_k(s)$ are the *latent conflict episode* and the *latent unrest episode*. m^C_{st} and m^U_{st} are everything we have now apart from the space-time fields: covariates, own and cross lags of recorded events (δ, κ and neighbour terms), country effects, cell effects and static Matérn fields. There are three space-time fields: v^S (shared), v^C (conflict only) and v^U (unrest only).

Dynamics of the three fields.

$$\begin{aligned}v^S_k(s) &= \varphi_S\, v^S_{k-1}(s) + \varepsilon^S_k(s),\\ v^C_k(s) &= \varphi_C\, v^C_{k-1}(s) + \psi_{U\to C}\, \bar{L}^U_{k-1}(s) + \varepsilon^C_k(s),\\ v^U_k(s) &= \varphi_U\, v^U_{k-1}(s) + \psi_{C\to U}\, \bar{L}^C_{k-1}(s) + \varepsilon^U_k(s),\end{aligned}$$

where $\bar{L}^U_{k-1}(s) = (1-\omega)\, L^U_{k-1}(s) + \omega\cdot$(mean of L^U_{k-1} over the neighbours of s), so last step's unrest episode here and next door feeds conflict now; likewise $\bar{L}^C$. The innovations ε^S_k, ε^C_k, ε^U_k are independent Matérn fields with sea barriers, independent over k, each with its own range and SD.

What each parameter answers.

- λ and v^S: *shared episodes*, which move both outcomes in the same step and place; φ_S is their persistence.
- $\psi_{U\to C}$: *latent escalation*. An unrest episode raises conflict risk one step later, here and next door, and the effect then decays at rate φ_C. $\psi_{C\to U}$ is the reverse direction.
- κ (in m^C): *short-run local escalation* from recorded events, as now.

Current models as special cases. Current *extended* model: $\psi_{U\to C} = \psi_{C\to U} = 0$ and $v^C \equiv 0$ (v^S is its shared field, v^U its unrest-only field). Current *joint* model: also $v^U \equiv 0$. Univariate space-time model of conflict: $\lambda = 0$, $v^U \equiv 0$, no unrest data. The main summary is the impulse response: the change in conflict risk over the next $1, 2, \ldots$ steps after a one-SD unrest episode, split into its shared and its escalation parts.

Why it fits our machinery. Each step of (v^S, v^C, v^U) is Gaussian given the previous step, so the joint precision is sparse (block bidiagonal in time, finite-volume sparsity in space). The Laplace engine, METIS ordering and case-control subsampling are unchanged. Stationarity requires the one-step transition of (v^S, v^C, v^U) to have spectral radius below one, checked at the estimate. Hyperparameters: 13 (against 9 in the extended model).

Identification. Shared episodes move both outcomes in the *same* step; escalation moves conflict *after* unrest. They are separated by timing, so the model needs enough steps: $\Delta = 4$ weeks on 1° cells for West Africa (91 steps, about 1.8×10^5 latent values) and $\Delta = 8$ on 2° cells for Africa and the Middle East (46 steps, about 1.3×10^5). Longer memory than one AR(1) can be added as a sum of a fast and a slow component, which stays sparse.

Simulation check (before any real-data claim). S2 design, 20 replicates per truth: (i) shared episodes only; (ii) latent escalation only ($\psi_{U\to C} = 0.3$); (iii) both; (iv) short-run κ only. Fit the new model, the current extended model and the univariate model. Criteria: bias and coverage of $\psi_{U\to C}$, λ and κ; no spurious ψ under (i) and (iv). *Go/no-go:* if ψ and λ are not separated in (i)–(iii), stop and fall back to regional λ and κ.

Cost. Implementation and tests: 1 day. Simulation: 1 day on the CRC. Real-data fits (two regions, warm-started, $r = 0.25$): 1–2 days. Rewriting methods and application: 1 day. Paper: the latent escalation model becomes the main model, with the current models as special cases; the application reports ψ, λ, κ and impulse responses.